

CLASS 9th
PHYSICS ADVANCE SHEET

A. Motion

1. A stone is thrown vertically upward with 20 m/s. Find max height and time to return.
2. Two cars moving in opposite directions with velocities 40 km/h and 60 km/h. Find relative velocity.

B. Force & Laws of Motion

1. A 50 kg man pushes a 100 kg box with 200 N force. Find acceleration & friction if $\mu = 0.2$
2. A bullet of mass 10 g hits a wall at 50 m/s and stops in 0.01 s. Find impulse.

C. Gravitation & Pressure

1. A satellite orbits Earth at radius 7000 km. Find orbital speed.
2. A cube of side 0.5 m submerged in water. Calculate upthrust.

D. Work, Energy & Power

1. A 5 kg body slides down a 10 m high slope. Find final velocity ignoring friction.
2. A machine does 500 J work in 10 s. Find power.

E. Optics & Waves

1. A concave mirror has focal length 15 cm. Object at 10 cm. Find image position.
2. Two waves of frequency 500 Hz and 0.68 m wavelength. Find speed.